

Brian Mboya

Senior Software Engineer

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Brian is a Research Fellow at Wolfram Institute. He studied computer science in Dedan Kimathi University of Technology in Kenya. He is a programmer with experience in building applications using Python, Rust and Javascript programming languages. He is interested in many things both academic and non academic. Some of the fields he likes studying and reading about include cryptography, number theory, game theory, cognitive and computational neuroscience. An area that he is particularly fascinated about is the human brain, and how it works. His main area of study is attention and memory. Looking for the synthesis between computation and neural networks. His current work explores the behavior of Cellular Automata and Turing machines to identify rules that produce complex behaviours in simple programs and reveal insights into intelligent attributes. He received summer school scholarships attending different summer schools, including Computational Neuroscience and Machine Learning in Africa 2023, Simons Computational Neuroscience Imbizo 2024 and Explore and create projects at the frontiers of science, technology and innovation, Wolfram Research 2024. Aside from the above mentioned, he loves poetry and has written a few poems that he thinks are interesting, with focus on the theme of death and consciousness.

Areas of Expertise

◆ Infrastructure Management ◆ Research & Development ◆ Software Testing & Debugging ◆ Academic Research ◆ Wolfram language ◆ Mathematica	◆ Software Design & Architecture ◆ System Maintenance & Updates ◆ Continuous Process Improvement	◆ Team Leadership & Training ◆ Technical Troubleshooting ◆ Cross-functional Collaboration
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Technical Proficiencies

Mathematica | Wolfram Language | Python | JavaScript | SQL | Docker | Linux | PostgreSQL | MySQL | ReactJs | Redux | Git | AWS | Agile Development | Memcached | Redis | RabbitMQ

Career Experience

Wolfram Institute, Remote
Research Fellow

Jan 2026 – Present

Duties will include:

- Computational research as directed by Stephen Wolfram
- Writing and/or assisting with the writing of papers to introduce work done by the Institute to new audiences
- Building tools (paclets, functions, other pre-approved ways of allowing the general public to interact with our work)
- Attend and actively participate, lead or contribute to virtual and in person events
- Provide services under the direction of Stephen Wolfram in pursuit of research and education goals defined by the Wolfram

Affiliate Researcher

Sep 2024 – Jan 2026

Work on a Click to Copy project that aims to modernise and enhance the accessibility of the code used to create the iconic figures from Stephen Wolfram's *A New Kind of Science (NKS)* using the Wolfram Language and Mathematica.

- Generate figures using computational methods and principles described in the book.
- Optimised code by using techniques in the Wolfram language to make it easy for researchers reading and using the book's code to copy and run the code in their notebooks.
- Parse graphical data through tabulars and geometrical transformation.

Soapbox Inc., Remote
Senior Software Engineer

April 2022 – Present

Lead all facets of storefront, order, inventory, warehouse, and shipping operations. Establish back-end algorithms to implement efficient React components for users. Utilise Docker for deployment of containerization process. Coordinate multiple third-party APIs and manage data received from all storefronts such as Amazon, Etsy, Shopify, and Walmart.

- Streamlined workflow by introducing new API integrations into the system that scaled to 20 daily warehouses with each warehouse handling over 1 million orders.
- Optimised performance by providing exceptional training to new junior engineers that saw an increase in productivity in development and feature acceptance by 30%.
- Maintained support for Shippo, UPS and Easypost carrier services like Fedex, UPS and DHLCommerce that increased the amount of new customers by 15%.

Software Engineer

July 2020 – April 2022

Worked on integrating 3PL Logistic packaging technology in a system that enabled faster and better ways for our customers to manage their warehouses. Provided DevOps support and maintenance by implementing new scripts to automate the workflow of running code on AWS service.

- Optimised system performance by refactoring existing code to use newer and faster technology. This was a tremendous boost in the performance of the system, by 40% faster than the initial system.

The-Bhub, Nairobi
Software Engineer

Jan 2020 – July 2020

Manage system configuration and maintain source code for both server as well as client side. Automate deployment process of source code whilst ensuring customers are updated daily about the health of the system through automated notification. This saw an increase in positive customer feedback and growth in the system's daily user visits to 1 million users.

- Conducted research and posted new programming and examined findings for publication.
- Gained in-depth understanding of algorithms through my research for building a scalable system that scaled based on users visits. This system scaled to 2 million daily users.

Andela, Nairobi
Software Engineer

Jan 2018 – Oct 2019

Provided mentorship to the new engineers and helped guide the whole code-base. Established CircleCI CI/CD pipeline for automating testing components and deployment process, whilst maintaining a steady release cycle. Collaborated with DevOps team for migrating product to Jenkins pipeline hosted on servers. Created and maintained Relational Databases (PostgreSQL, MySQL) on GCP/AWS by using terraform for IAC.

- Integrated with third-party APIs, processed data, mentored engineers to attain top performance levels, which saw an increase in faster data processing and low latency in the system.
- Achieved system-level environment isolation and prepared complete containers by deploying containerization through utilisation of Docker. This ensured seamless deployment of our system and growth in productivity by 27%.

Additional Experience

Software Engineer, Craftux, Nairobi
Volunteer, Trend in Africa, Nairobi

Education

Jun, 2024: Explore and create projects at the frontiers of science, technology and innovation | Wolfram Research

Jan, 2024: Summer Course in Simons Computational Neuroscience Imbizo | Imbizo.Africa

Jun, 2023: Summer Course in Computational Neuroscience and ML | TREND in Africa

2015 - 2018: B.Sc. in Computer Science | Dedan Kimathi University of Technology, Nyeri